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Structure-based lead optimization to improve potency and selectivity of a novel tetrahydroimidazo[1,2-*a*]pyridine-5-carboxylic acid series of heparanase-1 inhibitor

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ABSTRACT

Heparanase-1 (HPSE1) is an *endo*-β-D-glucuronidase that is the only mammalian enzyme known to cleave heparan sulfate (HS) of heparan sulfate proteoglycans (HSPG), a key component of the glycocalyx layer of the vascular endothelium matrix. Inhibition of HPSE1 has therapeutic potential for cancer and proteinuric kidney diseases. We previously reported that **2** showed a moderate potency as an HPSE1 inhibitor and an issue of selectivity against *exo*-β-D-glucuronidase (GUSβ) and glucocerebrosidase (GBA) remained. A structure-based lead optimization of **2** using X-ray co-crystal structure analysis and fragment molecular orbital calculation resulted in **4e**, which showed a more than 7-fold increase in HPSE1 inhibitory activity. The subsequent introduction of a methyl group into the 6-hydroxy group of **4e** resulted in **18** with reduced inhibitory activities against GUSβ and GBA while maintaining the inhibitory activity against HPSE1. The inhibitory activities of **18** against serum HPSE1 in mice were significant and lasted for 4 h at doses of 3, 30, and 100 mg/kg. Compound **18** could be a novel lead compound for HPSE1 inhibitors with improved inhibitory activity against HPSE1 and increased HPSE1 selectivity over GUSβ and GBA.

1. Introduction

Heparanase-1 (HPSE1, EC 3.2.1.166) is an *endo*-β-D-glucuronidase that is the only mammalian enzyme known to cleave heparan sulfate (HS) of heparan sulfate proteoglycans (HSPG), a key component of the glycocalyx layer of the vascular endothelium matrix.[1] HPSE1 is initially produced as an inactive pre-proenzyme and is cleaved by signal peptidase to pro-HPSE1, which is then processed by cathepsin L to active HPSE1, a heterodimer of an *N*-terminal 8 kDa subunit and a C-terminal 50 kDa subunit.[2,3,4]

The cleavage of HS by HPSE1, which leads to the remodeling of the extracellular matrix and releasing of growth factors and cytokines bound to HS, promotes both physiological and pathological processes such as angiogenesis, immune cell migration, inflammation, wound healing, and metastasis, so that HPSE1 should be deeply associated with cancer in diverse situations.[5] HPSE1 is also thought to be associated with several proteinuric kidney diseases such as diabetic nephropathy (DNP) and nephrotic syndrome. An increased glomerular HPSE1 expression and the following decreased HS in the glomerular basement (GBM) are commonly observed both in experimental models such as rat and mouse models of streptozotocin (STZ)-induced DNP, rat adriamycin nephropathy and passive Heymann nephritis models, and in human glomerular diseases such as DNP, minimal change nephrotic syndrome, and IgA nephropathy;[6] these suggest that the loss of HS in GBM contributes to

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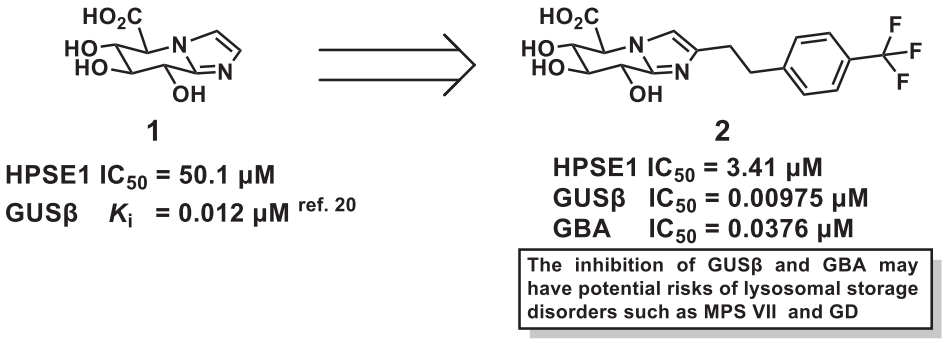


Fig. 1. The SAR and rational of optimization from **1** to **2**.[21.]

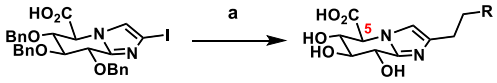
the development of proteinuria.[7] In contrast, glomerular HS was preserved and albuminuria was reduced by HPSE1 gene deficiency in experimental glomerulonephritis model mice induced by anti-GBM antibody or lipopolysaccharide.[8] In addition, the HPSE1 inhibition of a specific HPSE1 inhibitor roneparstat (SST0001), which mimics the structure of HS, decreased the extent of albuminuria and renal damage in diabetic model mice[9] and acute kidney injury model mice.[10] These findings suggest that the inhibition of HPSE1 has therapeutic potential for cancer and proteinuric kidney diseases.

To date, as the development of HS-mimic HPSE1 inhibitors, pixatimod (PG545)[11] has advanced to clinical development for the treatment of cancer combined with nivolumab (anti-programmed cell death 1 antibody) and cyclophosphamide.[12] Although muparfostat (PI-88)[13] and roneparstat[14] were also evaluated as HPSE1 inhibitor candidates in clinical trials for the treatment of cancer,[15,16] the administration route for these inhibitors was limited to injection because of a mixture of highly sulfated oligosaccharides. As for small molecules, 2,3-dihydro-1,3-dioxo-1*H*-isoindole-5-carboxylic acid,[17] a sugar analogue SF4,[18] and a glycoside CRM646-A[19] have been reported. In addition to these compounds, we previously reported that compound **1**, known as an *exo*-β-D-glucuronidase (GUSβ, EC:3.2.1.31) inhibitor,[20] also had a weak inhibitory activity against HPSE1 and compound **2** was designed and synthesized to improve the inhibitory activity against HPSE1 though a structure–activity relationship (SAR) study using the X-ray co-crystal structure of **1** and HPSE1[21] (Fig. 1).

Although **2** had a moderate potency as HPSE1 inhibitor (HPSE1 IC_{50} = 3.41 μM), the potency of which was expected to be higher, an issue of poor selectivity of **2** against GUSβ (IC_{50} = 0.00975 μM) and glucocerebrosidase (glucosylceramidase, GlcCerase, GBA, EC:3.2.1.45) (IC_{50} = 0.0376 μM) remained. GUSβ and GBA are lysosomal enzymes that play an essential role in the degradation of glycoproteins and glycolipids in lysosomes, and their deficiency is associated with several lysosomal storage disorders. Mucopolysaccharidosis type VII (MPS VII) is caused by the deficiency of GUSβ, which leads to partial degradation and the accumulation of glycosaminoglycans in many tissues throughout the body, with consequent cellular damage and organ dysfunction, resulting in hydrops fetalis, musculoskeletal deformities, neurodegeneration, and hepatosplenomegaly leading in most cases to premature death.[22,23,24] Gaucher disease (GD) is caused by the deficiency of GBA due to pathogenic biallelic variants in GBA1, which leads to the accumulation of glucocerebroside, resulting in a fatty substance, in the spleen, liver, bone marrow, and rarely in the lungs or central nervous system.[25,26] These facts implied that the inhibition of GUSβ or GBA may have potential risks to cause unfavorable side effects, hence we needed further studies not only to increase the inhibitory activity against HPSE1 but also to reduce the inhibitory activity against GUSβ and GBA to avoid these concerns.

2. Results and discussion

The syntheses of **4a**, **4b**, **4c**, **4d**, **4e**, **5a**, **5b**, and **5c** are shown in Scheme 1. A Sonogashira coupling reaction between **3**[21] and the corresponding arylacetylenes with Superstable Pd, CuI, and trimethylamine in MeCN at 80 °C gave alkyne derivatives, and then the obtained products were hydrogenated with $Pd(OH)_2$ in AcOEt, MeOH, H_2O, and AcOH under hydrogen atmosphere to yield **4a**–**4e** and **5a**–**5c**.

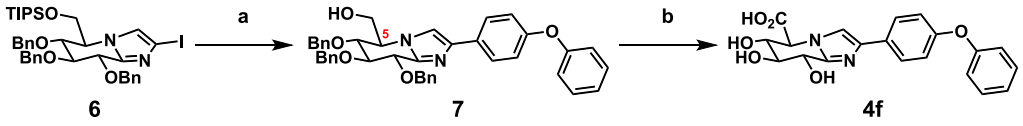


Scheme 1. Syntheses of **4a**, **4b**, **4c**, **4d**, **4e**, **5a**, **5b**, and **5c**. Reagents and conditions: (a) i) arylacetylene, Superstable Pd, CuI, Et_3N, MeCN, 80 °C; ii) H_2 gas, $Pd(OH)_2$, AcOEt, MeOH, H_2O, AcOH, rt, 34% (**4a**), 28% (**4b**), 22% (**4c**), 18% (**4d**), 35% (**4e**), 25% (**5a**), 8% (**5b**), 31% (**5c**) in 2 steps.

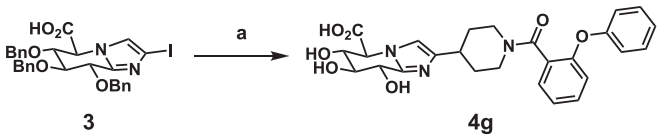
The synthesis of **4f** is shown in Scheme 2. A Suzuki coupling between triisopropyl silyl ether **6**[21] and the corresponding boronic acids with $PdCl_2(dppf)\bullet CHCl_3$ and sodium carbonate in dioxane and H_2O at 80 °C, and the following deprotection of TIPS group with TBAF afforded **7** with a 66% yield in 2 steps. Oxidation of **7** with Jones reagent converted the 5-hydroxymethyl group into a 5-carboxy group and finally the removal of all benzyl groups simultaneously using $Pd(OH)_2$ in AcOEt, MeOH, H_2O, and AcOH gave **4f** with a 24% yield in 2 steps.

The synthesis of **4g** is shown in Scheme 3. The synthetic route was similar to that of **4a** in Scheme 1, which initiated with carboxylic acid **3**. A Suzuki coupling between **3** and the corresponding boronic acids with $PdCl_2(dppf)\bullet CHCl_3$ and sodium carbonate in dioxane and H_2O at 80 °C, and the following removal of all benzyl groups and the reduction of alkenyl part using $Pd(OH)_2$ in AcOEt, MeOH, H_2O, and AcOH produced **4g** with a 12% yield.

The synthesis of **18** is shown in Scheme 4. First, 3,4-dihydroxy groups of commercially available starting material **8** were benzylated with sodium hydride and benzyl bromide in DMF followed by reaction with triethyl silane and trifluoroacetic acid (TFA) at room temperature so that selective removal of the benzyl acetal group at 4-position resulted in **9** with a 63% yield in 2 steps. Next, methylation of the hydroxy group with sodium hydride and iodomethane converted from **9** into **10** with a 92% yield. Selective hydrolysis of the methyl glucoside of **10** with hydrochloric acid was achieved using strontium chloride to yield secondary alcohol, followed by oxidation with tetrapropyl ammonium perruthenate (TPAP), *N*-methyl morpholine (NMO), and molecular sieve (MS) 4Å to convert into **11** with a 37 % yield in 2 steps. The lactone **11** was converted into lactam **12** with a 46 % yield in 3 steps including hydrolysis with ammonia solution in MeOH, Dess-Martin



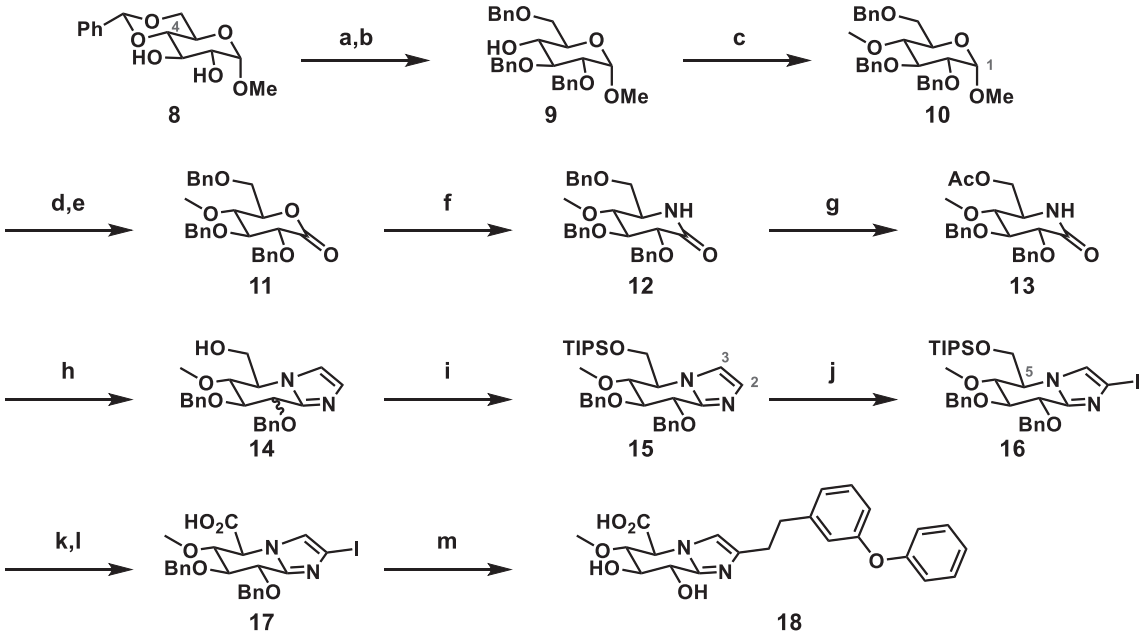
Scheme 2. Synthesis of **4f**. Reagents and conditions: (a) i) (4-phenoxyphenyl)boronic acid, $PdCl_2(dppf)\bullet CHCl_3$, Na_2CO_3, dioxane, H_2O, 80 °C; ii) TBAF, THF, 0 °C, 66% in 2 steps; (b) i) Jones reagent, acetone, rt; ii) H_2 gas, $Pd(OH)_2$, AcOEt, MeOH, H_2O, AcOH, rt, 24% in 2 steps.



Scheme 3. Synthesis of **4g**. Reagents and conditions: (a) i) (2-phenoxyphenyl)-[4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3,6-dihydro-2*H*-pyridin-1-yl] methanone, $PdCl_2(dppf)\bullet CHCl_3$, Na_2CO_3, dioxane, H_2O, 80 °C; ii) H_2 gas, $Pd(OH)_2$, AcOEt, MeOH, H_2O, AcOH, rt, 12% in 2 steps.

oxidation, and reduction of hemiaminal with boron trifluoride diethyl etherate and triethylsilane.[27] The benzyl group at the 6-hydroxymethyl position of **12** was selectively removed and the obtained alcohol was acetylated using H_2SO_4, AcOH, and Ac_2O to acquire **12** with a 73% yield. Thionation of glucolactam **13** with Lawesson's reagent provided a thionolactam, followed by condensation with aminoacetaldehyde dimethyl acetal and cyclisation of the resulting amidine in the presence of *p*-TsOH·H_2O in toluene and H_2O at 80 °C to produce tetrahydroimidazo[1,2-*a*]pyridine **14**,[27] with a 63% yield, as mixtures of diastereomers and simultaneously the acetyl group was removed. The hydroxy group of **14** in diastereomer mixtures was protected with triisopropylsilyl chloride and imidazole in DMF; an unnecessary diastereomer was then removed to yield **15** with a 58% yield. Di-iodination at the 2,3-positions of **15** with *N*-iodosuccinimide (NIS) at 80 °C and the following removal of the iodine at the 3-position with EtMgBr at −10 °C produced **16** with a 73% yield in 2 steps.[28] Removal of the triisopropylsilyl group at the C5-hydroxymethyl position of **16** with TBAF and the following Jones oxidation produced **17** with a 57% yield in 2 steps. Finally, A Sonogashira coupling between **17** and 1-ethynyl-3-phenoxybenzene in a similar manner described above was achieved with Superstable Pd, CuI, and trimethylamine in MeCN at 60 °C, followed by removal of all benzyl groups to yield **18** with a 40% yield in 2 steps.

To improve the HPSE1 inhibitory activity of **2**, we have performed a fragment molecular orbital (FMO) calculation to investigate the interaction between **2** and HPSE1 for potentiating the already observed interaction of **2** with HPSE1 rather than acquiring a new interaction with HPSE1. Jia Shi *et al* previously reported the FMO calculation results of HPSE1 and tetrasaccharides.[29] FMO calculation is a computational method, in which a large molecular system is divided into small fragments, e.g., each amino acid residue of a target protein is able to be represented by a fragment.[30,31,32,33] The FMO calculation results are considered reasonable to understand an interaction between a compound and each amino acid residue of a target protein by examining energetic components for each pair of fragments, called inter-fragment interaction energy (IFIE) or pair interaction energy (PIE).[34] Fig. 2**A** shows the X-ray co-crystal structure of **2** and HPSE1 (PDB code: 7YI7)[21] and the PIE values calculated for each amino acid of HPSE1 interacting with **2** observed in the X-ray co-crystal structure (Fig. 2**B**). The highest PIE values of the two glutamates Glu^{225} (PIE = −70.6 kcal/mol) and Glu^{343} (PIE = −57.5 kcal/mol) in the catalytic site of HPSE1,[35] which interact with the imidazole moiety of **2**, suggested their contributions to the interaction between **2** and HPSE1. In addition, the high PIE values of Asp^{62} (PIE = −48.1 kcal/mol) and Tyr^{391} (PIE = −26.4 kcal/mol) and the moderate PIE values of Thr^{97} (PIE = −15.1 kcal/mol) and the backbone NHs of Gly^{349} (PIE = −17.9 kcal/mol) and Gly^{350} (PIE =



Scheme 4. Synthesis of **18**. Reagents and conditions: (a) NaH, BnBr, DMF, 0 °C to rt, 94%; (b) Et_3SiH, TFA, $CHCl_3$, rt, 67%; (c) NaH, MeI, DMF, 0 °C, 92%; (d) $SrCl_3$, 6*N*-HClaq, AcOH, 80 °C, 45%; (e) tetrapropyl ammonium perruthenate (TPAP), *N*-methyl morpholine (NMO), molecular sieve (MS) 4Å, $CHCl_3$, rt, 83%; (f) i) 7*N*-ammonia in MeOH, MeOH, rt; ii) Dess-Martin reagent, $CHCl_3$, H_2O, 0 °C; iii) Et_3SiH, $BF_3\bullet Et_2O$, $CHCl_3$, −10 °C, 46% in 3 steps; (g) H_2SO_4, AcOH, Ac_2O, rt, 73%; (h) i) Lawesson's reagent, toluene, 80 °C; ii) aminoacetaldehyde dimethyl acetal, rt; iii) *p*-TsOH·H_2O, toluene, H_2O, 80 °C, 63% in 3 steps; (i) TIPSCl, imidazole, DMF, rt, 58%; (j) i) NIS, DMF, 80 °C; ii) EtMgBr, THF, −10 °C, 73% in 2 steps; (k) TBAF, THF, 0 °C, 95%; (l) Jones reagent, acetone, rt, 60%; (m) i) 1-ethynyl-3-phenoxybenzene, Superstable Pd, CuI, Et_3N, MeCN, 60 °C; ii) H_2 gas, Pd/C, AcOEt, rt, 40% in 2 steps.

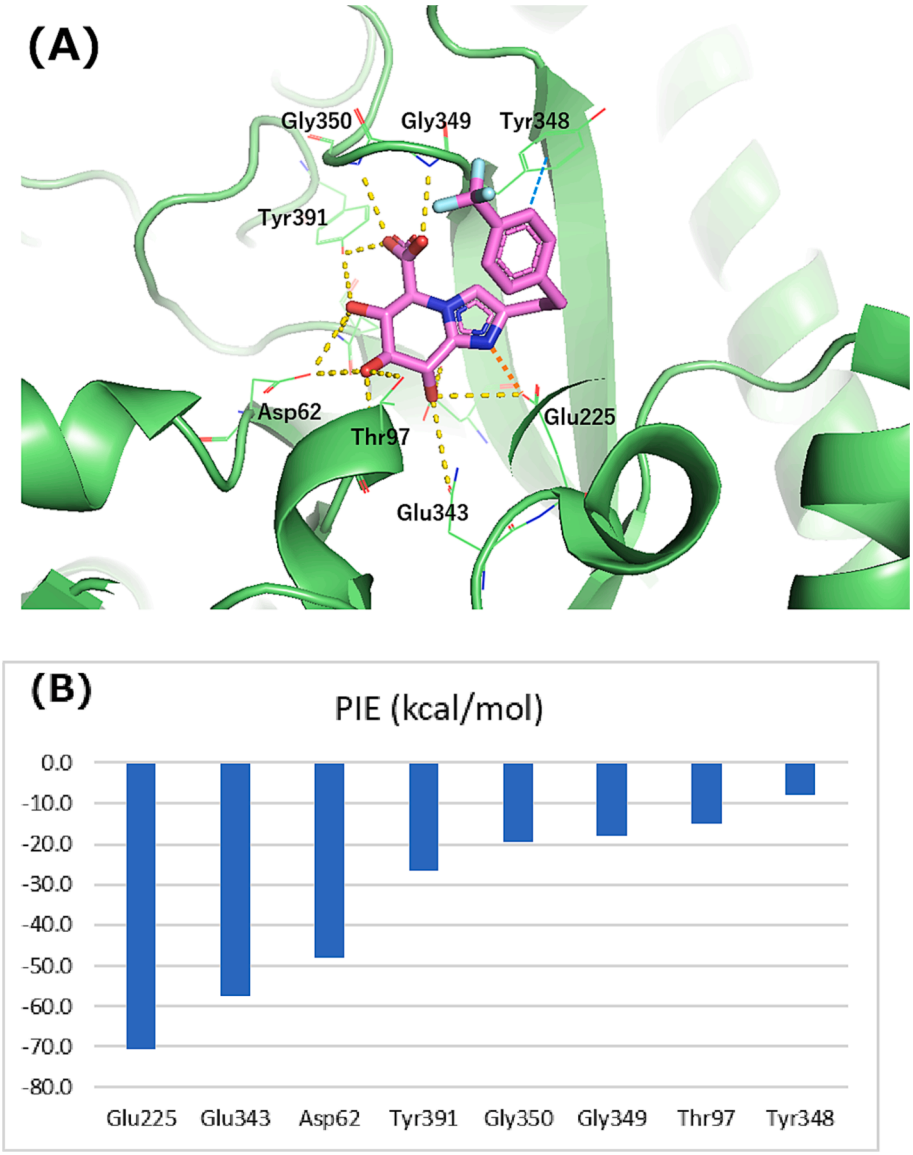


Fig. 2. (A) X-ray co-crystal structure of **2** (purple) and HPSE1 (PDB code: 7YI7),[21] highlighting key interactions of **2** with residues of HPSE1. The yellow dotted lines indicate a hydrogen bond interaction, the orange dotted line indicates an ionic interaction, and the cyan dotted line indicates a CH/π interaction. (B) The PIE values calculated for each amino acid of HPSE1 interacting with **2** observed by X-ray analysis.

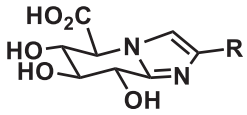
−19.4 kcal/mol), which interact with the 6,7-dihydroxy and 5-carboxylate of **2**, also suggested their contribution to the interactions between **2** and HPSE1. The results of the PIE values implied that the imidazole, 6,7-dihydroxy, and 5-carboxylate moieties of **2** were crucial for the inhibitory activity of **2** against HPSE1.

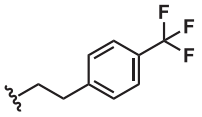
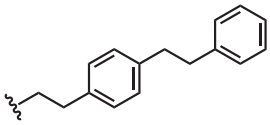
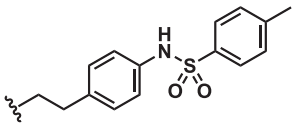
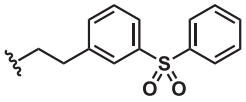
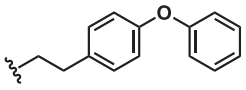
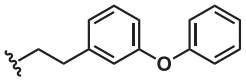
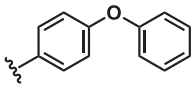
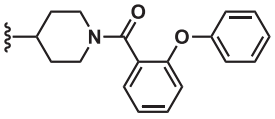
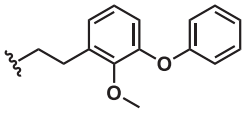
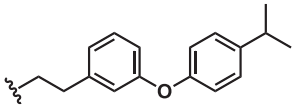
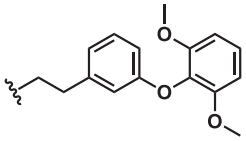
To further improve the inhibitory activity of **2** against HPSE1, we focused our attention on Tyr^{348} (PIE = −8.1 kcal/mol), which appeared to interact with the 4-trifluoromethyl benzene moiety of **2** through a CH/π interaction. Replacement of the 4-trifluoromethylphenethyl group with a variety of functional groups would be a feasible strategy by using convenient synthetic reactions such as Sonogashira and Suzuki coupling reactions. Therefore, we tried to replace the 4-trifluoromethylphenethyl group of **2** with a variety of functional groups including two aromatic rings to potentiate the interaction with Tyr^{348}.

Table 1 shows the series of compounds with a functional group examined. First, replacing 4-trifluoromethyl group of **2** with phenethyl group resulted in **4a** (IC_{50} = 2.00 μM), which had an approximately 1.7-fold increase in HPSE1 inhibitory activity compared with that of **2**. Next, the optimization of the linker between two phenyl groups of **4a** was performed. As a result, **4b** with a sulfonamide linker (IC_{50} = 1.34 μM) had a more potent HPSE1 inhibitory activity than that of **4a**. Compound **4c** with a sulfonyl linker (IC_{50} = 0.748 μM), in which the amino group was removed from the sulfonamide linker of **4b**, showed an approximately 2-fold increase in activity compared to that of **4b**. Replacing the sulfonyl linker of **4c** with an oxygen linker resulted in **4d** with a 4-phenoxyphenyl group (IC_{50} = 0.586 μM) and **4e** with a 3-phenoxy group (IC_{50} = 0.467 μM), which exhibited more potent inhibitory activity against HPSE1 than that of **4c**. Additionally, deletion of the ethylene linker between the phenoxyphenyl group and imidazole moiety of **4d** resulted in **4f** (IC_{50} = 1.10 μM) with less potent activity than that of **4d**. Compound **4g** with a 1-piperidinylcarbonyl linker at the *ortho*-position of the phenoxyphenyl group showed lower activity (IC_{50} = 1.75 μM) than those of **4d**, **4e**, and **4f**.

Since **4e** exhibited the most potent inhibitory activity about 7-fold higher than that of **2** through the exploration of an additional phenyl group, several analogues incorporating substituents into the middle or terminal benzene ring of **4e** were investigated. However, incorporating a methoxy group into the middle benzene ring (**5a**, IC_{50} = 1.60 μM), an isopropyl group into the terminal benzene ring (**5b**, IC_{50} = 1.38 μM), and two methoxy groups into the terminal benzene ring (**5c**, IC_{50} = 9.95

Table 1
HPSE1 inhibitory activities of compounds with modifications on the 4-trifluoromethylphenyl group of **2**.



Compound	R	IC_{50} (μM)
2		3.41
4a		2.00
4b		1.34
4c		0.748
4d		0.586
4e		0.467
4f		1.10
4g		1.75
5a		1.60
5b		1.38
5c		9.95

μM) lowered HPSE1 inhibitory activity compared to **4e**.

An X-ray analysis was performed on the complex of **4e** and HPSE1 to model the binding modes (Fig. 3**A**). Compound **4e** bound to the active site of HPSE1 with a similar binding mode as that of **2** (PDB code: 7YI7).[21] The imidazole moiety of **4e** was thought to interact with the two glutamate residues Glu^{225} and Glu^{343}, which are located in the catalytic site of HPSE1, via non-covalent interactions such as an electrostatic interaction. The hydrogen bond interactions between the 5-carboxylate of **4e** and the backbone NHs of Gly^{349} and Gly^{350} of HPSE1, as well as those between the 5-carboxylate and 6,7-dihydroxy groups of **4e** and Asp^{62}, Asn^{224}, Tyr^{391}, and Glu^{343} of HPSE1 were also observed.

We also performed an FMO calculation to investigate the interaction between **4e** and HPSE1, and then compared the PIE values, calculated for the amino acids of HPSE1 interacting with **2** and **4e** observed by X-ray analysis (Fig. 3**B**). The PIE values of several residues (Glu^{343}, Glu^{225}, Asp^{62}, Tyr^{391}, Gly^{349}, and Gly^{350}) of HPSE1 interacting with **4e** were similar to those interacting with **2**. Although a large decrease in the PIE value of Tyr^{348} was initially expected for **4e** with the improved inhibitory activity against HPSE1 by increasing the number of benzene rings, the PIE value of Tyr^{348} (PIE = −11.0 kcal/mol) decreased only slightly compared to that interacting with **2** (PIE = −8.1 kcal/mol). This was thought to be because the terminal phenoxy group of **4e** and the residue of Tyr^{348} contribute to the CH/π interaction, but the middle benzene part of **4e** can hardly participate in the interaction due to its distance and angle from Tyr^{348}. Conversely, larger decrease in the PIE value of Arg^{272} [PIE = −11.4 kcal/mol (**4e**), 4.7 kcal/mol (**2**)] of HPSE1 was observed by changing the compounds from **2** to **4e**. X-ray analysis of the complex of **4e** and HPSE1 (Fig. 3**A**) indicates that the residue of Arg^{272} is involved in a cation–π interaction with the terminal 3-phenoxy part of **4e**. Although the middle benzene part of **4e** is not expected to strengthen the interaction with the residue of Tyr^{348}, it seems to acquire the weak interaction with the residue of Gln^{270} through a dispersion interaction, so that the moderate decrease in the PIE values of Gln^{270} [PIE = −3.8 kcal/mol (**4e**), 0.5 kcal/mol (**2**)] of HPSE1 (see supplementary data, Table S2). In addition, the PIE values of Glu^{343} and Asn^{224} were also decreased. We assumed that this was attributed to the interaction of the 3-phenoxyphenethyl group of **4e** with the residues of HPSE1 such as Gln^{270}, Arg^{272}, and Tyr^{348}, which made the structural fluctuation of the tetrahydroimidazo[1,2-*a*]pyridine core of **4e** more stable, thus enhancing the interaction of **4e** with Glu^{343} and Asn^{224}.

Replacement of the 4-trifluoromethyl group of **2** with a 3-phenoxy group (**4e**) led to a more than 7-fold increase in the inhibitory activity against HPSE1; however, issues regarding the lack of selectivity against GUSβ (IC_{50} = 0.00773 μM) and GBA (IC_{50} = 0.0528 μM) remained.

To efficiently improve the HPSE1 selectivity of **4e** against GUSβ and GBA while maintaining the activity against HPSE1, we investigated the different features of the active sites of HPSE1, GUSβ, and GBA by comparing the X-ray co-crystal structures of these enzymes interacting with **4e**. HPSE1 is an *endo*-β-D-glucuronidase that prefers to cleave an internal sugar linkage of HS rather than a terminal sugar linkage of HS, whereas GUSβ is an *exo*-β-D-glucuronidase that prefers to cleave a terminal sugar linkage rather than an internal sugar linkage, which allowed us to expect the difference of the amino acid residues surrounding the enzymatic pockets of HPSE1 and GUSβ interacting with **4e**.

X-ray analysis of the complex of **4e** and HPSE1 revealed the presence of an empty space around the 6-hydroxy group of **4e** as shown in Fig. 4**A** (brown colored area). Actually, the X-ray co-crystal structure of an oligomer dp4 and HPSE1 (PDB code: 5E9C), reported by Liang Wu *et al*, showed that the *N*-sulfo-D-glucosamine-6-sulfate moiety of the oligomer dp4 was located in the empty space,[36] suggesting a promising approach to introduce a substituent at the 6-position of **4e**. The 6-hydroxy group of **4e** was found to interact with Gln^{383} and Tyr^{391} through hydrogen bond interactions, in which the 6-oxygen atom of **4e** could work as a hydrogen bond acceptor even if a functional group such as a methyl group is introduced into the 6-hydroxy group of **4e**.

Regarding GUSβ, when the structure of **4e** was superposed on the X-ray co-crystal structure of a sugar analogue, 2-propyl uronic isofagomine, and GUSβ (PDB code: 6LEJ)[37] (Fig. 4**B**), the 4-hydroxy group of the sugar analogue, corresponding to the 6-hydroxy group of **4e**, seemed to interact with the residues of Trp^{549} and Asp^{163} of GUSβ through hydrogen bond interactions, in which the 4-hydroxy group appears to work as a hydrogen bond donor. The results suggested that replacement of the 4-hydroxy group of the sugar analogue would cause the loss of the hydrogen bond interactions. In addition, Phe^{554} of GUSβ was located very close to the 4-hydroxy group of the sugar analogue, so that the introduction of a functional group such as methyl group into the

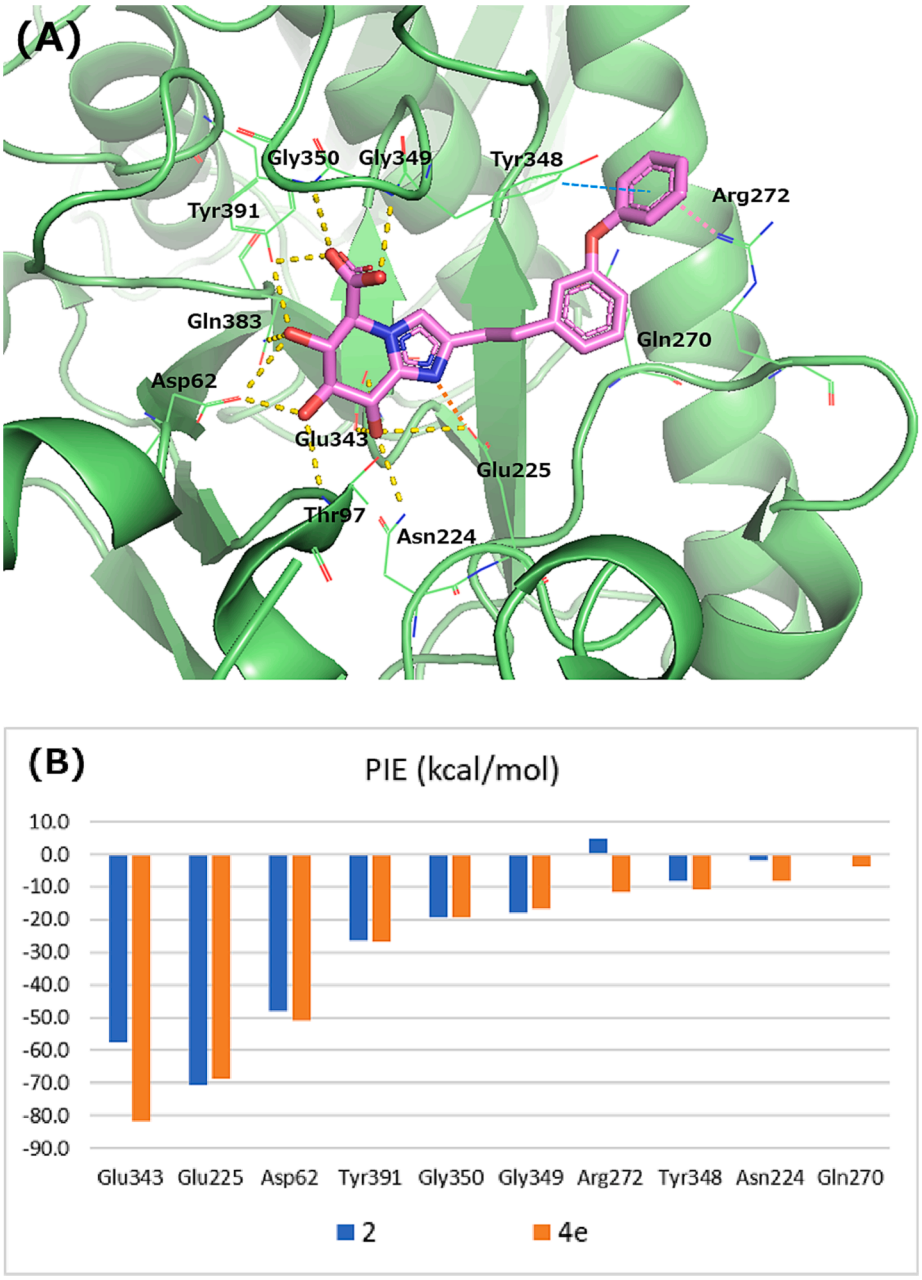


Fig. 3. (A) X-ray co-crystal structure of **4e** (purple) and HPSE1 (PDB code: 8JYG), highlighting key interactions of **4e** with residues of HPSE1. The yellow dotted lines indicate a hydrogen bond interaction, the orange dotted line indicates an ionic interaction, the cyan dotted line indicates a CH/π interaction, the pink dotted line indicates a cation-π interaction. (B) Comparison of the PIE values calculated for each amino acid of HPSE1 interacting with **2** or **4e** observed by X-ray analysis.

4-hydroxy group of the sugar analogue would cause steric hindrance with Phe[554]. These findings allowed us to expect that methylation of the 6-hydroxy group of **4e** would impair the interaction between **4e** and GUSβ.

Concerning GBA as well as GUSβ, when the structure of **4e** was superposed on the X-ray co-crystal structure of a sugar analogue, *N*-butyldeoxynojirimycin, and GBA (PDB code: 2V3D)[38] (Fig. 4**C**), the 5-hydroxy group of the sugar analogue, which would correspond to the 6-hydroxy group of **4e**, seemed to interact with Trp[381]and Asp[127] of GBA through hydrogen bond interactions, in which the 5-hydroxy group appears to work as a hydrogen bond donor. As in the case of GUSβ, Phe[128] of GBA was located very close to the 5-hydroxy group of the sugar analogue, so that the introduction of a functional group into the 5-hydroxy group of the sugar analogue or the 6-hydroxy group of **4e** would not only cause the loss of the hydrogen bond interactions but also cause steric hindrance with Phe[128], resulting in the attenuation of GBA inhibitory activity.

These considerations urged us to design **18**, in which a methyl group was introduced into the 6-hydroxy group of **4e** (Fig. 5). Compound **18** was expected to show decreased inhibitory activities against GUSβ and GBA due to the loss of the hydrogen bond interactions as hydrogen bond donor and the steric hindrance with Phe[554] or Phe[128], while maintaining the inhibitory activity against HPSE1.

As expected, the methylation of the 6-hydroxy group of **4e**, compound **18** was effective in reducing the inhibitory activities against GUSβ and GBA while maintaining the inhibitory activity against HPSE1. The profiles comparing **18** with **4e** were shown in Table 2. The inhibitory activity of **18** (IC_{50} = 0.442 μM) against HPSE1 was almost similar to that of **4e**, whereas **18** [IC_{50} = 0.839 μM (GUSβ), 0.439 μM (GBA)] was more than 108-fold and 8-fold less potent in inhibiting GUSβ and

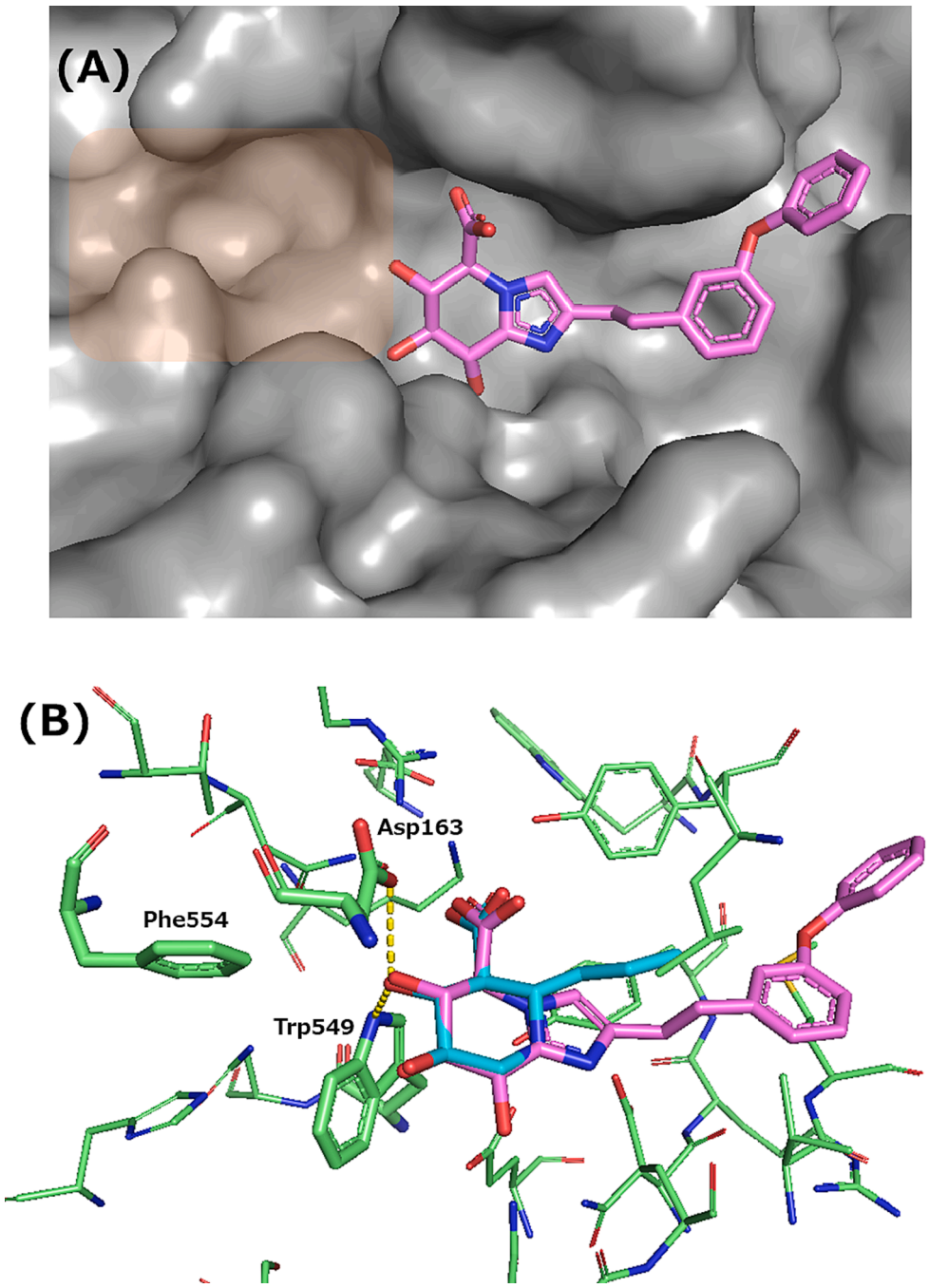


Fig. 4. (A) An empty space (brown) around the 6-hydroxy group of **4e** (purple) in the X-ray co-crystal structure of **4e** and HPSE1 (PDB code: 8JYG). (B) The superposition of **4e** (purple) on the X-ray co-crystal structure of a sugar analogue C6-propyl uronic isofagomine (cyan) and GUSβ (PDB code: 6LEJ),[37] highlighting key interactions of the sugar analogue with residues of GUSβ. The yellow dotted lines indicate a hydrogen bond interaction. (C) The superposition of **4e** (purple) on the X-ray co-crystal structure of a sugar analogue *N*-butyl-deoxynojirimycin (cyan) and GBA (PDB code: 2V3D),[38] highlighting key interactions of the sugar analogue with residues of GBA. The yellow dotted lines indicate a hydrogen bond interaction.

GBA than **4e**, respectively. Regarding the physicochemical property of **18**, the calculated Log D values at pH 7.4 (Clog $D_{7.4}$) were appropriate (Clog $D_{7.4} = -0.92$), which falls within the range ($-1.5 <$ Clog $D_{7.4} <$ 1.5) of the "Drug-Like Property" Rules for carboxylate-containing oral drug candidates previously reported by Alexander Böcker *et al.*[39] As for the pharmacokinetic (PK) profiles of **18** in mice, the serum protein binding was high (95.5%), comparable to that of **4e** (96.4%) because of the carboxyl group of **18**, which generally afford serum albumin binding affinity by interacting with its arginine residue. In addition, **18** had low total plasma clearance (CL_{total} = 350 mL/h/kg) and a volume of distribution at steady state (Vd_{ss} = 666 mL/kg), which were roughly comparable to those of **4e** (CL_{total} = 491 mL/h/kg, Vd_{ss} = 439 mL/kg) and the terminal half-life of **18** ($T_{1/2}$ = 2.1 h) was similar to that of **4e** ($T_{1/2}$ = 1.6 h). The urinary excretion rate of **18** accounted for a substantial fraction of the dose (31.1%) and was a slightly lower than that of **4e** (44.2%), which may be due to a slight increase in the lipophilicity of **18** by methyl incorporation into the hydroxy group of **4e**. Collectively, these results indicated that **18** was a physiochemically and pharmacokinetically favorable compound with improved selectivity towards HPSE1 and GUSβ/GBA.

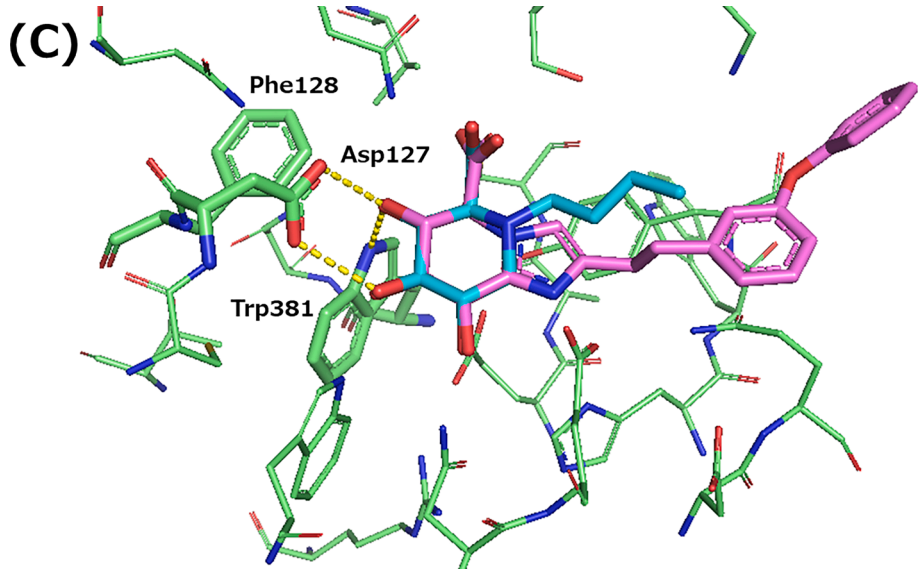


Fig. 4. (*continued*).

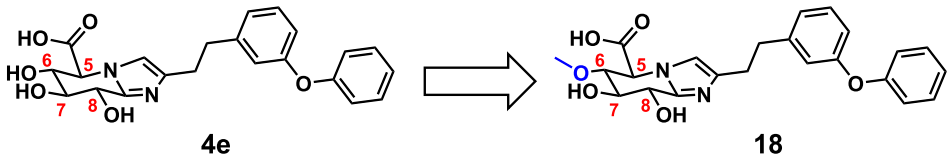
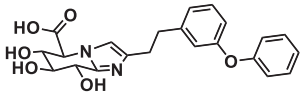
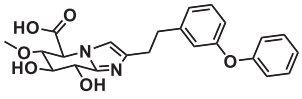


Fig 5. A design strategy to improve the selectivity towards HPSE1 and GUSβ/GBA by introducing a methyl group into the 6-hydroxy group of tetrahydroimidazo[1,2-*a*]pyridine core.

Table 2
Pharmacological, physicochemical, and pharmacokinetic profiles of **4e** and **18.**

	 **4e**	 **18**
HPSE1 IC_{50} (μM)	0.467	0.442
GUSβ IC_{50} (μM)	0.00773	0.839
GBA IC_{50} (μM)	0.0528	0.439
CLog $D_{7.4}$[a]	−1.5	−0.92
Protein Binding (%)	96.4	95.5
CL_{total} (mL/h/kg)	491	350
Vd_{ss} (mL/kg)	439	666
$T_{1/2}$ (h)	1.6	2.1
Urinary Excretion Rate (%)	44.2	31.1

[a] CLog $D_{7.4}$ values of **4e** and **18** were calculated using ACD/Percepta™ software (Advanced Chemistry Development, Inc.).

Finally, we investigated the serum HPSE1inhibitory activity of **18** in mice (Fig. 6). Compound **18** was administered subcutaneously (s.c.) at doses of 3, 30, and 100 mg/kg suspended in 10 w/v% SBE-β-CD solution to C57BL/6J mice, followed by measurement of serum HPSE1 activity 1 and 4 h after administration. The inhibitory effect of **18** against serum HPSE1 activity was significant 1 h after s.c. administration at all doses and the effect lasted for 4 h.

We also examined the plasma concentrations of **18** in mice 1 and 4 h after s.c. administration at 3, 30, and 100 mg/kg of **18** to investigate PK-pharmacodynamics (PD) relationship between the plasma concentration and the HPSE1 inhibitory activity of **18** (Table 3). The increase in plasma concentrations of **18** was observed according to the increase in the doses of **18**. All the mean plasma concentrations of **18** in mice (1240, 9820, 24200, 191, 1070, and 1880 ng/mL) showing the significant HPSE1 inhibitory activity of **18** were higher than the median inhibition concentration of **18** (IC_{50} = 188 ng/mL), calculated from the molecular weight of **18** (424.45 g/mol) and the IC_{50} value of **18** against HPSE1 (0.442 μmol/L). These results indicated that serum HPSE1 inhibitory activity in mice would be used to evaluate therapeutic potential of compounds in mouse cancer and proteinuric kidney diseases models.

3. Conclusion

In summary, we performed the structure-based lead optimization of **2** which we previously reported, not only to increase the inhibitory activity against HPSE1 but also to reduce the inhibitory activities against GUSβ and GBA, resulting in **18**, which showed an improved inhibitory activity against HPSE1 and an increased HPSE1selectivity over GUSβ

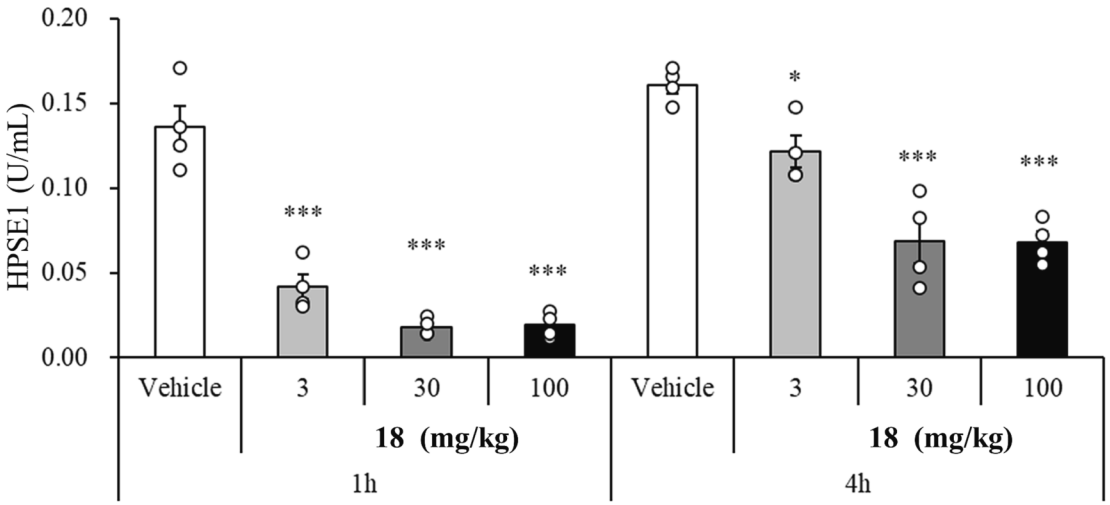


Fig. 6. Serum HPSE1 activities at 1 and 4 h after s.c. administration of vehicle or 3, 30, 100 mg/kg of **18** on mice.Mean ± standard error (n = 4 per group). *: $p < 0.05$ vs. vehicle, ***: $p < 0.001$ vs. vehicle (Dunnett's test).

Table 3
Plasma concentrations at 1 and 4 h after s.c. administration of 3, 30, 100 mg/kg of **18** on mice.

Plasma concentrations of 18 (ng/mL)		
Dose (mg/kg)	1 h	4 h
3	1240 ± 250	191 ± 64
30	9820 ± 2070	1070 ± 700
0	24200 ± 400	1880 ± 390

Mean ± standard deviation (n = 4 per group).

and GBA.

First, based on the result of the X-ray co-crystal structure analysis of **2** and HPSE1 and the FMO analysis of the molecular interaction of **2** with HPSE1, replacement of the 4-trifluoromethylphenethyl group of **2** with functional groups including two aromatic rings was investigated to potentiate the interaction between **2** and Tyr^{348}, resulting in **4e** with 3-phenoxyphenethyl group, which showed a more than 7-fold increase in HPSE1 inhibitory activity compared with that of **2**. The FMO analysis of **4e** suggested that the interaction of the 3-phenoxyphenethyl group with the residues of HPSE1 would cause the structural fluctuation of the tetrahydroimidazo[1,2-*a*]pyridine core of **4e** more stable, and the further interaction enhancement of the **4e** with Glu^{343} and Asn^{224}.

Next, we designed a compound to reduce the inhibitory activities of **4e** against GUSβ and GBA while maintaining the inhibitory activity against HPSE1 by overlaying **4e** with the GUSβ or GBA -bound conformation of sugar analogues in the X-ray co-crystal structures. The introduction of a methyl group into the 6-hydroxy group of **4e** resulted in **18**, whose inhibitory activity against HPSE1 was very similar to that of **4e**, whereas GUSβ and GBA inhibitory activities of **18** was greater than 108-fold and greater than 8-fold less potent than those of **4e**, respectively.

Finally, we investigated the serum HPSE1 inhibitory activity of **18** in mice as a potent HPSE1 inhibitor. The inhibitory effect of **18** against serum HPSE1 activity was significant and lasted for 4 h at all doses of 3, 30, and 100 mg/kg. The PK-PD relationship for **18** showed that all the plasma concentrations of **18** exhibiting the significant inhibitory effect against serum HPSE1 activity were higher than the median inhibition concentration of **18**.

Compound **18** could be a novel lead compound for potent HPSE1 inhibitors, with improved inhibitory activity against HPSE1 and increased HPSE1 selectivity over GUSβ and GBA. Further studies such as the introduction of a bulkier functional group to the 6-methoxy group of **18** are warranted to improve the HPSE1 selectivity over GUSβ and GBA.

4. Experimental section

4.1. Chemistry

4.1.1. General experimental information

All reagents and solvents were purchased from commercial suppliers and were used without purification or were prepared according to published procedures.

The ^{1}H NMR and ^{13}C NMR spectra of compounds synthesized in this study were recorded using an Avance III HD 400 (Bruker Corp., Billerica, MA, USA) or JNM-ECA500, JNM-ECA600 (JEOL Ltd., Tokyo, Japan) spectrometer, and the chemical shifts were expressed in δ (:) values, with tetramethylsilane used as the internal standard (s = singlet, d = doublet, t = triplet, q = quartet, m = multiplet, brs = broad singlet, br d = broad doublet, br dd = broad double doublet, and br t = broad triplet). Mass spectra were recorded on a Shimadzu LCMS-2010EV (Shimadzu Corp., Kyoto, Japan) or Micromass Platform LC (Micromass Ltd., Manchester, UK). High-resolution (HR) mass spectral data were acquired using an LCMS-IT-TOF equipped with an electrospray ionization (ESI)/atmospheric pressure chemical ionization (APCI) dual ion source (Shimadzu Corp.).

Several compounds were purified using reverse-phase preparative HPLC [Agilent 1260 Infinity/Agilent 6130 (Agilent Technologies Inc., Santa Clara, CA, USA), GX-281, UV/VIS-155, 331 PUMP, 332 PUMP, SOFTA Model 300S ELSD (Gilson Inc., Middleton, WI, USA)] under the following conditions: column, Sunfire prep C18 OBD (5.0 μm, 30 mm × 50 mm) (Waters Corp., Milford, MA, USA), YMC-Actus Triart C18 (5.0 μm, 30 mm × 50 mm) (YMC Co., Ltd., Kyoto, Japan), Xbridge Prep C18 OBD (5.0 μm, 30 mm × 50 mm) (Waters Corp.), or XSelect CSH C18 (5.0 μm, 30 mm × 50 mm) (Waters Corp.); flow, 50 mL/min; linear gradient, 10%−95% acetonitrile in water containing 0.1% formic acid for 7.5–11.5 min; detection wavelength, 254 nm.

The purity of the synthesized compounds was determined using an LC-MS system (Agilent 1290 Infinity, Agilent Technologies Inc.) under the following conditions: column, ACQUITY UPLC CSH C18 (1.7 μm, 2.1 × 50 mm) (Waters Corp.); flow, 0.8 mL/min; linear gradient, 20%–99% acetonitrile in water containing 0.1% formic acid for 1.2 min. The detection wavelengths were 210 nm and 254 nm. All final compounds were confirmed to have a purity of ≥ 95% at 254 nm.

4.1.2. Syntheses of key compounds *4e* and *18*

Syntheses of the other compounds are shown in Supplementary data.

*Synthesis of (5S,6R,7S,8S)-6,7,8-trihydroxy-2-[2-(3-phenoxyphenyl) ethyl]-5,6,7,8-tetrahydroimidazo[1,2-a]pyridine-5-carboxylic acid **4e***

A solution of **3** (300 mg, 0.492 mmol), copper iodide (18.7 mg, 0.0983 mmol) Superstable Palladium (0) (208 mg, 0.0983 mmol), and trimethylamine (0.342 mL, 2.46 mmol) in acetonitrile (7 mL) was heated to 60 °C and 1-ethynyl-3-phenoxybenzene (286 mg, 1.47 mmol) in acetonitrile (5 mL) was added in a dropwise manner. The solution was then stirred at 60 °C for 1 h. After cooling to room temperature, the reaction mixture was diluted with ethyl acetate and washed with saturated ammonium chloride solution and brine in that order. The organic layer was dried over anhydrous magnesium sulfate, filtered and the filtrate was concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 0%–15% methanol in chloroform) to obtain benzyl (5*S*,6*R*,7*S*,8*S*)-6,7,8-tribenzyloxy-2-[2-(3-phenoxyphenyl)ethynyl]-5,6,7,8-tetrahydroimidazo[1,2-*a*]pyridine-5-carboxylic acid (282 mg, 0.417 mmol, 85%) as a pale-brown amorphous substance.

Palladium hydroxide (20 wt%) on activated carbon (564 mg, 4.01 mmol) was added to a solution of (5*S*,6*R*,7*S*,8*S*)-6,7,8-tribenzyloxy-2-[2-(3-phenoxyphenyl)ethynyl]-5,6,7,8-tetrahydroimidazo[1,2-*a*]pyridine-5-carboxylic acid (282 mg, 0.417 mmol) in acetic acid (5.0 mL), ethyl acetate (3.0 mL), methanol (1.0 mL), and water (1.0 mL) under a nitrogen atmosphere. The flask was exposed to flowing hydrogen gas, and the mixture was vigorously stirred overnight. The flask was then exposed to flowing nitrogen gas, and the reaction mixture was filtered, and the filtrate was concentrated under reduced pressure. The residue was purified using reverse-phase preparative HPLC and slurred with isopropyl ether to obtain **4e** (70 mg, 0.171 mmol, 41%) as a colorless solid.

^{1}H NMR (400 MHz, DMSO-d_6) δ ppm 2.66–2.74 (m, 2H) 2.81–2.90 (m, 2H) 3.66 (br t, J = 6.1 Hz, 1H) 3.92 (br t, J = 6.1 Hz, 1H) 4.18–4.23 (m, 1H) 4.30–4.37 (m, 1H) 5.29–5.37 (m, 1H) 6.77–6.85 (m, 2H) 6.91 (br s, 1H) 6.97–7.06 (m, 3H) 7.13 (br t, J = 7.2 Hz, 1H) 7.29 (t, J = 7.2 Hz, 1H) 7.36–7.43 (m, 2H).

^{13}C NMR (126 MHz, DMSO-d_6) δ ppm 29.72, 34.91, 63.63, 67.69, 69.84, 73.05, 115.12, 115.92, 118.46, 118.59, 123.30, 123.41, 129.80, 130.01, 139.37, 144.26, 144.35, 156.61, 156.68, 170.33.

MS (ESI): *m/z* calcd for $C_{22}H_{22}N_2O_6$ $[M+H]^+$: 411.15, found 411.1, $[M-H]^-$: 409.15, found 409.2.

HRMS (ESI/APCI dual): *m/z* calcd for $C_{22}H_{22}N_2O_6$ $[M+Na]^+$: 433.1370, found 433.1409, $[M-H]^-$: 409.1405, found: 409.1425.

*Synthesis of (5S,6R,7R,8S)-7,8-dihydroxy-6-methoxy-2-[2-(3-phenoxyphenyl)ethyl]-5,6,7,8-tetrahydroimidazo[1,2-a]pyridine-5-carboxylic acid **18** from **8** in 19 steps*

*(2R,3R,4S,5R,6S)-4,5-dibenzyloxy-2-(benzyloxymethyl)-6-methoxy-tetrahydropyran-3-ol **9***

Triethylsilane (53.3 mL, 334 mmol) and trifluoroacetic acid (25.5 mL, 334 mmol) were added to a solution of **8** (38.6 mg 83.5 mmol) in $CHCl_3$ (278 mL) at 0 °C in an ice bath, and the mixture was stirred for 6 h at room temperature. The reaction mixture was quenched with sat. Na_2CO_3 aq. and extracted with $CHCl_3$. The organic layer was separated using ISOLUTE® Phase Separator and concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 5%–34% ethyl acetate in hexane) to obtain **9** (26.0 g, 56.0 mmol, 67%) as a colorless oil.

^{1}H NMR (400 MHz, Chloroform-d) δ ppm 3.38 (s, 3H) 3.50–3.56 (m, 1H) 3.56–3.63 (m, 1H) 3.66–3.71 (m, 2H) 3.74–3.81 (m, 1H) 4.12 (q, J = 7.1 Hz, 1H) 4.51–4.61 (m, 2H) 4.62–4.69 (m, 2H) 4.75 (t, J = 11.5 Hz, 2H) 5.00 (d, J = 11.5 Hz, 1H) 7.23–7.47 (m, 15H).

*(2R,3R,4S,5R,6S)-4,5-dibenzyloxy-2-(benzyloxymethyl)-3,6-dimethoxy-tetrahydropyran **10***

60% Sodium hydride in oil (1.72 g, 43.1 mmol) was added to a solution of **9** (10 g, 21.5 mmol) in DMF (108 mL) at 0 °C cooling in an ice bath. After stirred for 15 min, iodomethane (2.68 mL, 43.1 mmol) was added to the mixture cooled in an ice bath, and the mixture was stirred at 0 °C for 1 h. The reaction mixture was quenched with sat. NH_4Cl aq. and extracted with a mixture of EtOAc-hexane (2:1, 200 mL). The organic layer was washed twice with water followed by dried over anhydrous magnesium sulfate, filtered and concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 10%–50% ethyl acetate in hexane) to obtain **10** (9.52 g, 19.9 mmol, 91%) as a colorless oil.

^{1}H NMR (400 MHz, CHLOROFORM-d) δ ppm 3.29–3.35 (m, 1H) 3.37 (s, 3H) 3.46 (s, 3H) 3.50 (br dd, J = 9.2, 3.4 Hz, 1H) 3.60–3.69 (m, 3H) 3.86 (t, J = 9.2 Hz, 1H) 4.49–4.67 (m, 4H) 4.74–4.83 (m, 2H) 4.93 (d, J = 10.9 Hz, 1H) 7.26–7.40 (m, 15H).

MS (ESI): *m/z* calcd for $C_{29}H_{34}O_6$ $[M+Na]^+$: 501.24, found 501.3, $[2M+Na]^+$: 979.48, found 979.5.

*(3R,4S,5R,6R)-3,4-dibenzyloxy-6-(benzyloxymethyl)-5-methoxy-tetrahydropyran-2-one **11***

6*N*-HCl aq. (9.95 mL, 19.9 mmol) and strontium chloride hexahydrate (530 mg, 1.99 mmol) were added to a solution of **10** (9.52 g, 19.9 mmol) in AcOH (66.3 mL) at room temperature. The mixture was heated at 80 °C for 3 h. After cooling to room temperature, the reaction mixture was quenched with 1*N*-NaOH aq. and extracted with $CHCl_3$. The organic layer was separated using ISOLUTE® Phase Separator and concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 33%–50% ethyl acetate in hexane) to obtain (3*R*,4*S*,5*R*,6*R*)-3,4-dibenzyloxy-6-(benzyloxymethyl)-5-methoxy-tetrahydropyran-2-ol (4.17 g, 8.98 mmol, 45%) as a colorless oil as diastereomer mixtures.

^{1}H NMR (400 MHz, Chloroform-d) δ ppm 3.26–3.39 (m, 1H) 3.47 (s, 3H) 3.50–3.58 (m, 1H) 3.61–3.75 (m, 2H) 3.85 (br t, J = 9.2 Hz, 1H) 3.90–3.96 (m, 1H) 4.49–4.95 (m, 6H) 5.17–5.24 (m, 1H) 7.20–7.45 (m, 15H).

MS (ESI): *m/z* calcd for $C_{28}H_{32}O_6$ $[M+Na]^+$: 487.22, found 487.3, $[2M+Na]^+$: 951.44, found 951.4.

Molecular sieves 4 Å (4.17 g) and 4-methylmorpholine *N*-oxide (2.42 g, 20.6 mmol) were added to a solution of (3*R*,4*S*,5*R*,6*R*)-3,4-dibenzyloxy-6-(benzyloxymethyl)-5-methoxy-tetrahydropyran-2-ol (4.17 g, 8.98 mmol) in $CHCl_3$ (29.9 mL) at room temperature. After stirring for 30 min, tetrapropylammonium perruthenate (315 mg, 0.898 mmol) was added to the mixture at 0 °C in an ice bath and the mixture was stirred at 0 °C for 30 min followed by stirring at room temperature for 2 h. The reaction mixture was filtered, and the filtrate was concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 10%–33% ethyl acetate in hexane) to obtain **11** (3.45 g, 7.46 mmol, 83%) as a colorless oil.

^{1}H NMR (400 MHz, Chloroform-d) δ ppm 3.43 (s, 3H) 3.65–3.85 (m, 4H) 4.07–4.12 (m, 1H) 4.36–4.44 (m, 1H) 4.51–4.65 (m, 4H) 4.69–4.75 (m, 1H) 4.97 (d, J = 11.5 Hz, 1H) 7.22–7.41 (m, 15H).

MS (ESI): *m/z* calcd for $C_{28}H_{30}O_6$ $[M+Na]^+$: 485.20, found 485.2, $[2M+Na]^+$: 947.40, found 947.4.

*(3R,4S,5R,6R)-3,4-dibenzyloxy-6-(benzyloxymethyl)-5-methoxy-piperidin-2-one **12***

7*N*-ammonia in MeOH (2.7 mL, 18.9 mmol) was added to a solution of **11** (3.45 g, 7.46 mmol) in MeOH (10.7 mL) at room temperature and stirred for 2 h. The reaction mixture was concentrated under reduced pressure. The residue was azetroped with ethyl acetate twice to obtain (2*R*,3*S*,4*R*,5*R*)-2,3,6-tribenzyloxy-5-hydroxy-4-methoxy-hexanamide (3.30 g, 6.88 mol, 92%).

^{1}H NMR (400 MHz, Chloroform-d) δ ppm 3.44 (s, 3H) 3.54–3.70 (m, 3H) 3.86 (br d, J = 4.2 Hz, 1H) 3.98–4.06 (m, 1H) 4.19 (br d, J = 4.2 Hz, 1H) 4.50–4.74 (m, 6H) 5.40 (br s, 1H) 6.58 (br s, 1H) 7.25–7.39 (m, 15H).

MS (ESI): *m/z* calcd for $C_{28}H_{33}NO_6$ $[M+H]^+$: 480.23, found 480.3, $[M+Na]^+$: 502.23, found 502.3, $[2M+Na]^+$: 981.46, found 981.4.

The Dess-Martin periodinane (4.88 g, 11.5 mmol) and water (0.117 mL) were added to a solution of (2*R*,3*S*,4*R*,5*R*)-2,3,6-tribenzyloxy-5-hydroxy-4-methoxy-hexanamide (3.30 g, 6.88 mol) in $CHCl_3$ (36.0 mL) cooled with ice bath and the mixture was stirred at room temperature overnight. The reaction mixture was quenched with sat. $NaHCO_3$ aq. and $Na_2S_2O_3$ aq. sequentially followed by extraction with $CHCl_3$. The organic layer was separated using ISOLUTE® Phase Separator and

concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 50%–80% ethyl acetate in hexane) to obtain (3*R*,4*S*,5*S*)-3,4-dibenzyloxy-6-(benzyloxymethyl)-6-hydroxy-5-methoxy-piperidin-2-one (2.00 g, 4.19 mmol, 58%) as a pale-yellow oil as diastereomer mixtures.

^{1}H NMR (400 MHz, Chloroform-d) δ ppm 3.28–3.38 (m, 2H) 3.43–3.68 (m, 5H) 3.79–4.03 (m, 1H) 4.09–4.18 (m, 1H) 4.54–4.88 (m, 5H) 5.13 (br dd, J = 17.5, 11.4 Hz, 1H) 6.17–6.34 (m, 1H) 7.26–7.44 (m, 15H).

MS (ESI): *m/z* calcd for $C_{28}H_{31}NO_6$ $[M+Na]^+$: 500.22, found 500.3, $[2M+Na]^+$: 977.44, found 977.4.

Triethylsilane (2.27 mL, 14.2 mmol) was added to a solution of (3*R*,4*S*,5*S*)-3,4-dibenzyloxy-6-(benzyloxymethyl)-6-hydroxy-5-methoxy-piperidin-2-one (2.00 g, 4.19 mmol) in $CHCl_3$ at − 10 °C. After stirring for 30 min, boron trifluoride diethyl etherate (0.634 mL, 5.03 mmol) was added to the mixture at − 10 °C and stirred at − 10 °C for 1 h. The reaction mixture was quenched with sat. NH_4Cl aq. followed by extracted with $CHCl_3$. The organic layer was separated using ISOLUTE® Phase Separator and concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 50%–80% ethyl acetate in hexane) to obtain **12** (1.60 g, 3.47 mmol, 83%) as a colorless solid with a 46% yield in 3 steps from **11**.

^{1}H NMR (400 MHz, Chloroform-d) δ ppm 3.21–3.30 (m, 1H) 3.40–3.46 (m, 1H) 3.48 (s, 3H) 3.50–3.57 (m, 1H) 3.71 (br d, J = 8.9 Hz, 1H) 3.81 (t, J = 7.8 Hz, 1H) 3.96 (d, J = 7.8 Hz, 1H) 4.48–4.59 (m, 2H) 4.68–4.86 (m, 3H) 5.14 (d, J = 11.3 Hz, 1H) 5.92 (br s, 1H) 7.28–7.42 (m, 15H).

MS (ESI): *m/z* calcd for $C_{28}H_{31}NO_5$ $[M+Na]^+$: 484.22, found 484.3, $[2M+Na]^+$: 945.44, found 945.4.

[(2R,3R,4S,5R)-4,5-dibenzyloxy-3-methoxy-6-oxo-2-piperidyl]methyl acetate ***13***

Sulfuric acid (0.185 mL, 3.47 mmol) diluted with acetic acid (3.47 mL, 60.6 mmol) was added dropwise to a solution of **12** (1.60 g, 3.47 mmol) in acetic anhydride (41.9 mL, 443 mmol) at 0 °C in an ice bath, and the mixture was stirred for 3 h at room temperature. The reaction mixture was diluted with ethyl acetate and quenched with sat. $NaHCO_3$ aq. cooled in an ice bath. The organic layer was separated and dried over anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 50%–80% ethyl acetate in hexane) to obtain **13** (1.05 g, 2.54 mmol, 73%) as a pale-yellow solid.

^{1}H NMR (400 MHz, Chloroform-d) δ ppm 2.10 (s, 3H) 3.30 (t, J = 8.6 Hz, 1H) 3.53 (s, 3H) 3.56 (br s, 1H) 3.82 (t, J = 7.8 Hz, 1H) 3.94–4.04 (m, 2H) 4.43 (br d, J = 11.4 Hz, 1H) 4.68–4.84 (m, 3H) 5.12 (d, J = 11.4 Hz, 1H) 5.93 (br s, 1H) 7.31 (br s, 8H) 7.38–7.44 (m, 2H).

MS (ESI): *m/z* calcd for $C_{23}H_{27}NO_6$ $[M+H]^+$: 414.18, found 414.3, $[M+Na]^+$: 436.18, found 436.2, $[2M+Na]^+$: 849.36, found 849.4.

[(5R,6R,7S)-7,8-dibenzyloxy-6-methoxy-5,6,7,8-tetrahydroimidazo [1,2-a]pyridin-5-yl]methanol ***14***

Lawesson's reagent (770 mg, 1.90 mmol) was added to a solution of **13** (1.05 g, 2.54 mmol) in toluene (12.7 mL), and the mixture was stirred at 80 °C for 3 h. The reaction mixture was concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 25%–50% ethyl acetate in hexane) to obtain [(2*R*,3*R*,4*S*,5*R*)-4,5-dibenzyloxy-3-methoxy-6-thioxo-2-piperidyl]methyl acetate (1.20 g) as a green oil.

Next, [(2*R*,3*R*,4*S*,5*R*)-4,5-dibenzyloxy-3-methoxy-6-thioxo-2-piperidyl]methyl acetate (1.20 g) was dissolved in aminoacetaldehyde dimethyl acetal (7.14 mL, 69.8 mmol) and stirred overnight. The reaction mixture was diluted with ethyl acetate and washed with sat. NH_4Cl aq. and brine. The organic layer was separated and dried over anhydrous sodium sulfate, filtered, and concentrated under reduced pressure. The residue (1.10 g), which included [(2*R*,3*R*,4*S*,6*Z*)-4,5-dibenzyloxy-6-(2,2-dimethoxyethylimino)-3-methoxy-2-piperidyl]methyl acetate, was obtained as a colorless solid.

Finally, the residue (1.10 g) was dissolved in toluene (11 mL) and *p*-Toluenesulfonic acid monohydrate (1.04 g, 5.78 mmol) and water (1.10 mL) was added to the solution at room temperature. The mixture was heated at 80 °C for 6 h followed by stirring at room temperature overnight. The reaction mixture was diluted with ethyl acetate and quenched with sat. $NaHCO_3$ aq. The organic layer was separated, dried over anhydrous sodium sulfate, filtered, and concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 0%–10% methanol in chloroform) to obtain **14** (630 mg, 1.60 mmol, 63% in 3 staps) as a pale-yellow amorphous substance.

^{1}H NMR (400 MHz, Chloroform-d) δ ppm 3.52–3.68 (m, 3H) 3.74–3.94 (m, 1H) 4.15 (br s, 4H) 4.56–4.81 (m, 4H) 4.85–5.16 (m, 1H) 7.04–7.17 (m, 2H) 7.21–7.36 (m, 8H) 7.37–7.45 (m, 2H).

MS (ESI): *m/z* calcd for $C_{23}H_{26}N_2O_4$ $[M+H]^+$: 395.19, found 395.2.

[(5R,6R,7S,8S)-7,8-dibenzyloxy-6-methoxy-5,6,7,8-tetrahydroimidazo [1,2-a]pyridin-5-yl]methoxy-triisopropyl-silane ***15***

Triisopropylsilyl chloride (370 mg, 1.92 mmol) and imidazole (163 mg, 2.40 mmol) were added to a solution of **14** (630 mg, 1.60 mmol) in DMF (8.0 mL) at room temperature. The mixture was stirred at room temperature for 60 h. The reaction mixture was quenched with water followed by extraction with ethyl acetate-hexane (1:1). The organic layer was separated, washed twice with water, dried over anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 33%–50% ethyl acetate in hexane) to obtain **15** (510 mg, 0.926 mmol, 58%) as a yellow oil.

^{1}H NMR (400 MHz, Chloroform-d) δ ppm 1.00–1.10 (m, 21H) 3.55 (s, 3H) 3.60–3.65 (m, 1H) 4.00–4.18 (m, 4H) 4.65–4.73 (m, 2H) 4.75–4.82 (m, 1H) 4.89 (d, J = 11.7 Hz, 1H) 5.18 (d, J = 11.7 Hz, 1H) 7.11 (s, 1H) 7.13–7.16 (m, 1H) 7.25–7.35 (m, 8H) 7.44 (br d, J = 7.2 Hz, 2H).

MS (ESI): *m/z* calcd for $C_{32}H_{46}N_2O_4Si$ $[M+H]^+$: 551.32, found 551.4.

[(5R,6R,7S,8S)-7,8-dibenzyloxy-2-iodo-6-methoxy-5,6,7,8-tetrahydroimidazo[1,2-a]pyridin-5-yl]methoxy-triisopropyl-silane ***16***

N-Iodosuccinimide (2.08 g, 9.26 mmol) was added to a solution of **15** (510 mg, 0.926 mmol) in DMF at room temperature. The mixture was heated at 80 °C for 3 h. After cooling to room temperature, the reaction mixture was diluted with ethyl acetate-hexane (1;1) and quenched with sat. $NaHCO_3$ aq. and sat. $Na_2S_2O_3$ aq. sequentially. The organic layer was separated, washed twice with water, dried with anhydrous magnesium sulfate, filtered, and concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 20% ethyl acetate in hexane) to obtain [(5*R*,6*R*,7*S*,8*S*)-7,8-dibenzyloxy-2,3-diiodo-6-methoxy-5,6,7,8-tetrahydroimidazo[1,2-*a*]pyridin-5-yl] methoxy-triisopropyl-silane (640 mg, 0.797 mmol, 86%) as a colorless oil.

^{1}H NMR (400 MHz, Chloroform-d) δ ppm 0.91–1.13 (m, 21H) 3.42 (s, 3H) 3.93 (br d, J = 6.5 Hz, 2H) 4.04 (br s, 1H) 4.07–4.20 (m, 2H) 4.45–4.53 (m, 1H) 4.54–4.66 (m, 2H) 4.86 (br d, J = 12.2 Hz, 1H) 5.03 (br d, J = 12.2 Hz, 1H) 7.21–7.34 (m, 8H) 7.43 (br d, J = 7.1 Hz, 2H).

MS (ESI): *m/z* calcd for $C_{32}H_{44}I_2N_2O_4Si$ $[M+H]^+$: 803.12, found 803.2.

3 M Ethylmagnesium bromide in diethyl ether (0.400 mL, 1.20 mmol) was added to a solution of [(5*R*,6*R*,7*S*,8*S*)-7,8-dibenzyloxy-2,3-diiodo-6-methoxy-5,6,7,8-tetrahydroimidazo[1,2-*a*]pyridin-5-yl] methoxy-triisopropyl-silane (640 mg, 0.797 mmol) in THF (5.32 mL) at − 15 °C and the mixture was stirred at the same temperature for 30 min. The reaction mixture was quenched with sat. NH_4Cl aq. followed by extraction with ethyl acetate. The organic layer was dried with anhydrous magnesium sulfate, filtered and the filtrate was concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 20% ethyl acetate in hexane) to obtain **16** (460 mg, 0.680 mmol, 85%) as a yellow oil.

^{1}H NMR (400 MHz, Chloroform-d) δ ppm 0.96–1.17 (m, 21H) 3.51 (s, 3H) 3.55–3.64 (m, 1H) 3.96–4.05 (m, 2H) 4.05–4.15 (m, 2H) 4.58–4.68 (m, 2H) 4.70–4.76 (m, 1H) 4.83 (d, J = 11.7 Hz, 1H) 5.12 (d, J = 11.7 Hz, 1H) 7.21 (s, 1H) 7.24–7.35 (m, 8H) 7.41 (br d, J = 7.2 Hz,

2H).

MS (ESI): *m/z* calcd for $C_{32}H_{45}IN_2O_4Si$ $[M+H]^+$: 677.22, found 677.3.

(5S,6R,7S,8S)-7,8-dibenzyloxy-2-iodo-6-methoxy-5,6,7,8-tetrahydroimidazo[1,2-a]pyridine-5-carboxylic acid ***17***

1 M Tetrabutylammonium fluoride (1.02 mL, 1.02 mmol) was added to a solution of **16** (460 mg, 0.680 mmol) in THF (4.53 mL) cooled with ice bath and the mixture was stirred for 1 h. The reaction mixture was quenched with sat. NH_4Cl aq. followed by extraction with ethyl acetate. The organic layer was separated, dried with anhydrous magnesium sulfate, filtered, and the filtrate was concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 33–66% ethyl acetate in hexane) to obtain [(5*R*,6*R*,7*S*,8*S*)-7,8-dibenzyloxy-2-iodo-6-methoxy-5,6,7,8-tetrahydroimidazo[1,2-*a*]pyridin-5-yl]methanol (335 mg, 0.644 mmol, 95%) as a yellow oil.

^{1}H NMR (400 MHz, Chloroform-d) δ ppm 3.46–3.55 (m, 3H) 3.59 (br t, $J = 6.4$ Hz, 1H) 3.92–4.15 (m, 4H) 4.58–4.75 (m, 3H) 4.81 (br d, $J = 11.7$ Hz, 1H) 5.07 (d, $J = 11.7$ Hz, 1H) 7.18 (s, 1H) 7.24–7.36 (m, 8H) 7.37–7.44 (m, 2H).

MS (ESI): *m/z* calcd for $C_{23}H_{25}IN_2O_4$ $[M+H]^+$: 521.09, found 521.2.

0.7 M Jones reagent (14.7 mL, 10.3 mmol) was added to a solution of [(5*R*,6*R*,7*S*,8*S*)-7,8-dibenzyloxy-2-iodo-6-methoxy-5,6,7,8-tetrahydroimidazo[1,2-*a*]pyridin-5-yl]methanol (335 mg, 0.644 mmol) in acetone (53.6 mL) at room temperature and the mixture was stirred overnight. The reaction mixture was quenched with isopropyl alcohol and concentrated under reduced pressure. The residue was dissolved with ethyl acetate and washed with water and brine sequentially. The organic layer was separated, dried over magnesium sulfate, filtered, and the filtrate was concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 0–20% methanol in chloroform) to obtain **17** (205 mg, 0.384 mmol, 60%) as a colorless solid.

^{1}H NMR (400 MHz, Chloroform-d) δ ppm 3.42 (s, 3H) 4.04 (br s, 1H) 4.28 (br s, 1H) 4.38–4.52 (m, 2H) 4.66–4.80 (m, 3H) 4.95 (br d, $J = 12.0$ Hz, 1H) 6.93 (br s, 1H) 7.14 (br d, $J = 6.1$ Hz, 2H) 7.26–7.37 (m, 8H).

MS (ESI): *m/z* calcd for $C_{23}H_{23}IN_2O_5$ $[M+H]^+$: 535.07, found 535.2, $[M-H]^-$: 533.07, found 533.1.

(5S,6R,7R,8S)-7,8-dihydroxy-6-methoxy-2-[2-(3-phenoxyphenyl) ethyl]-5,6,7,8-tetrahydroimidazo[1,2-a]pyridine-5-carboxylic acid ***18***.

1-Ethynyl-3-phenoxy-benzene (158 mg, 0.814 mmol) in acetonitrile (2 mL) was added dropwise to a solution of **17** (145 mg, 0.271 mmol), copper iodide (10.3 mg, 0.0543 mmol) Superstable Palladium (0) (115 mg, 0.0543 mmol), and trimethylamine (0.189 mL, 1.36 mmol) in acetonitrile (4 mL). The mixture was heated at 60 °C for 1 h. After cooling to room temperature, the reaction mixture was diluted with ethyl acetate and washed with sat. NH_4Cl aq. and brine sequentially. The organic layer was separated, dried over anhydrous magnesium sulfate, filtered, and the filtrate was concentrated under reduced pressure. The residue was purified using silica gel column chromatography (OH, 0%–15% methanol in chloroform) to obtain benzyl (5*S*,6*R*,7*S*,8*S*)-7,8-dibenzyloxy-6-methoxy-2-[2-(3-phenoxyphenyl)ethynyl]-5,6,7,8-tetrahydroimidazo[1,2-*a*]pyridine-5-carboxylic acid (123 mg, 0.205 mmol, 75%) as a pale-brown amorphous substance.

^{1}H NMR (400 MHz, DMSO-d_6) δ ppm 3.38 (s, 3H) 4.13–4.20 (m, 1H) 4.35–4.41 (m, 1H) 4.48–4.53 (m, 1H) 4.60 (s, 2H) 4.74–4.81 (m, 1H) 4.86–4.93 (m, 1H) 5.09–5.25 (m, 1H) 7.04–7.10 (m, 4H) 7.24–7.35 (m, 11*H*) 7.40–7.45 (m, 4H) 7.52 (s, 1H).

MS (ESI): *m/z* calcd for $C_{37}H_{32}N_2O_6$ $[M+H]^+$: 601.23, found 601.3, $[M-H]^-$: 599.23, found 599.3.

Palladium hydroxide (20 wt%) on activated carbon (246 mg, 1.75 mmol) was added to a solution of (5*S*,6*R*,7*S*,8*S*)-7,8-dibenzyloxy-6-methoxy-2-[2-(3-phenoxyphenyl)ethynyl]-5,6,7,8-tetrahydroimidazo [1,2-*a*]pyridine-5-carboxylic acid (123 mg, 0.205 mmol) in acetic acid (2.5 mL), ethyl acetate (1.5 mL), methanol (0.5 mL), and water (0.5 mL) under a nitrogen atmosphere. The flask was exposed to flowing hydrogen gas, and the mixture was vigorously stirred overnight. The flask was then exposed to flowing nitrogen gas, and the reaction mixture was filtered, and the filtrate was concentrated under reduced pressure. The residue was purified using reverse-phase preparative HPLC to obtain **18** (46 mg, 0.108 mmol, 53%) as a colorless solid.

^{1}H NMR (400 MHz, DMSO-d_6) δ ppm 2.70–2.76 (m, 2H) 2.81–2.89 (m, 2H) 3.36–3.40 (m, 3H) 3.92 (br s, 1H) 3.97 (br s, 1H) 4.40 (br s, 1H) 4.78 (br s, 1H) 5.12–5.30 (m, 1H) 6.76 (br s, 1H) 6.80 (br d, $J = 7.8$ Hz, 1H) 6.92 (br s, 1H) 6.97–7.09 (m, 3H) 7.13 (br t, $J = 7.5$ Hz, 1H) 7.29 (br t, $J = 7.8$ Hz, 1H) 7.39 (br t, $J = 7.5$ Hz, 2H).

^{13}C NMR (126 MHz, DMSO-d_6) δ ppm 29.30, 34.75, 57.55, 59.52, 66.65, 71.01, 79.36, 115.51, 115.98, 118.47, 118.58, 123.31, 123.42, 129.81, 130.00, 138.79, 144.05, 144.13, 156.62, 156.67, 169.44.

MS (ESI): *m/z* calcd for $C_{23}H_{24}N_2O_6$ $[M+H]^+$: 425.16, found 425.2, $[M-H]^-$: 423.16, found 423.1.

HRMS (ESI/APCI dual): *m/z* calcd for $C_{23}H_{24}N_2O_6$ $[M+H]^+$: 425.1707, found 425.1717, $[M+Na]^+$: 447.1527, found 447.1503, $[M-H]^-$: 423.1562, found: 423.1570.

4.2. In vitro assays

4.2.1. Assay of human heparanase-1 (HPSE1) inhibition

The study of the human HPSE1 inhibition was performed using homogeneous time resolved fluorescence (HTRF) technology (PerkinElmer Japan Co., Ltd., Kanagawa, Japan). First, a recombinant human HPSE1 protein was diluted to 8 ng/mL with an enzyme buffer (20 mM Tris-HCl, 150 mM NaCl, 0.02% bovine serum albumin, 0.02% Pluronic F-127, pH7.5), followed by various concentrations of compounds (0.2 μL in DMSO) were added to 10 μL of the enzyme solutions, and the solutions were incubated for 10 min at room temperature. Next, 10 μL of biotin-heparan sulfate-Eu cryptate (0.8 μg/mL in 40 mM of acetate buffer, pH5.2) was added to the solution. After 60 min of incubation at 21 °C, 20 μL of stop solution (0.625 μg/mL Streptavidin-d2, 0.8 M potassium fluoride, 2 mg/mL heparin, 0.1 M phosphate buffer, pH8.0) was added to the reaction mixture and incubated for 15 min. Finally, the fluorescence (665 nm and 615 nm) was measured using ARVO (PerkinElmer Japan Co., Ltd., Kanagawa, Japan). The fluorescence ratio for 665 nm over 615 nm was then used to calculate the IC_{50} values.

4.2.2. Assay of human exo-β-D-glucuronidase (GUSβ)

The study of the human GUSβ inhibition was performed based on a previously described method.[40] The optimal substrate concentration and pH were determined in preliminary experiments. First, a recombinant human GUSβ protein was diluted with acetate buffer (100 mM, pH3.5), followed by addition of various concentrations of compounds to the enzyme solution, and the solution was incubated for 10 min at 37 °C. Next, 4-methylumbelliferyl-β-D-glucuronide (4-MUG) was added to the solution (1.5 μg/mL GUSβ, 50 μM 4-MUG, and 1% DMSO at a final concentration) and incubated at 37 °C for 60 min. Finally, the fluorescence (ex: 320 nm, em: 445 nm) was measured using EnVision (PerkinElmer Japan Co., Ltd., Kanagawa, Japan). The intensity of fluorescence was converted to product concentration using a standard curve prepared from 4-methylumbelliferone, and the IC_{50} values were calculated using the product concentrations.

4.2.3. Assay of human glucosylceramidase (GBA)

A citric acid solution (100 mmol/L) was mixed with disodium hydrogenphosphate solution (200 mmol/L) to prepare citric acid–Na_2HPO_4 buffer (pH5.5). An assay buffer was prepared by adding sodium taurocholate (final concentration of 0.2%) and Triton X-100 (final concentration of 0.1%) to citric acid–Na_2HPO_4 buffer. Recombinant human GBA was diluted with the assay buffer. Various concentrations of compounds were then added to the GBA solution, and the mixtures were pre-incubated for 10 min at 21 °C. Then, 4-methylumbelliferyl-β-D-glucopyranoside (4-MUBDG) was added to the mixtures (0.25 μg/mL GBA, 300 μM 4-MUBDG, and 1% DMSO at a final concentration). After 24 h of incubation at 21 °C, the fluorescence (ex: 320 nm, em: 445 nm) was

measured using EnVision (PerkinElmer Japan Co., Ltd., Kanagawa, Japan). The intensity of fluorescence was converted to product concentration using a standard curve prepared from 4-methylumbelliferone, and the IC_{50} values were calculated using the product concentrations.

4.3. In vivo study

All in vivo experimental protocols used in this study were approved by the Taisho Pharmaceutical Animal Care Committee and were in accordance with the Guidelines for Proper Conduct of Animal Experiments (Science Council of Japan, 2006).

Animals were maintained under controlled temperature (23 ± 3 °C) and humidity (50 ± 20%) conditions and a 12-h light/dark cycle (lights on at 07:15 h).

4.4. Inhibitory activity of **18** against serum HPSE1 in mice

Male C57BL/6J mice were purchased from CLEA Japan Inc. (Tokyo, Japan). Eight-week-old mice were randomly assigned to vehicle or **18** treatment (n = 4 per group). The mice were treated with vehicle or **18** at the dose of 3, 30, or 100 mg/kg by subcutaneous injection. After 1 or 4 h, blood samples were collected from the posterior vena cava under general anesthesia by isoflurane inhalation and centrifuged to obtain serum or plasma samples. The serum heparanase activities were assessed by Heparan Degrading Enzyme Assay Kit (TaKaRa, Cat. #MK412, Takara Bio, Siga, Japan) according to the manufacturer's instructions.

Plasma samples were mixed with organic solvent containing an internal standard and centrifuged. Aliquots of the supernatant were subjected to bioanalysis using a liquid chromatography–tandem mass spectrometry (LC-MS/MS). The instrument used was API4000 mass spectrometer (AB SCIEX, Framingham, MA) equipped with a high-performance liquid chromatography system (Shimadzu Corp., Kyoto, Japan).

4.5. Pharmacokinetic study

4.5.1. Pharmacokinetic analysis

The test compounds were administered intravenously by cassette dosing to fasted male C57BL/6J mice (n = 3) at a dose of 0.5 mg/kg each. Blood samples were collected into tubes containing ethylenediamine-*N*,*N*,*N*,*N*-tetraacetic acid dipotassium salt dihydrate at 0.0833, 0.25, 0.5, 1, 2, 4, 8, and 24 h after administration and centrifuged to obtain plasma samples. Urine samples were collected for 24 h after administration. Plasma or urine samples were mixed with organic solvent containing an internal standard and centrifuged. Aliquots of the supernatant were subjected to bioanalysis using an LC-MS/MS. The plasma concentration–time profiles were analyzed by a non-compartmental analysis with pharmacokinetic analysis software Phoenix WinNonlin 7.0 or 8.0 (Certara, Princeton, NJ). The urinary excretions were calculated as the percent of the dose recovered in urine over 24 h. The instrument used was QTRAP 5500 or Triple Quad 5500 mass spectrometer (AB SCIEX) equipped with a high-performance liquid chromatography system (Shimadzu Corp.).

4.5.2. In vitro serum protein binding

The protein binding assay was conducted using a 96-well equilibrium dialysis method. The pooled serum of male C57BL/6J mice was obtained from The Jackson Laboratory Japan, Inc. (Kanagawa, Japan), Aliquots of the DMSO solution of the compounds were diluted in the serum to yield a final concentration of 1 μg/mL. Serum samples were dialyzed with a dialysis membrane (molecular weight cut-off: 12000–14000 Da) against 50 mM sodium phosphate buffer at 37 °C for 4 h in a 5% CO_2-incubator. The concentration of the test compound on each side of the membrane was determined by LC-MS/MS. Serum protein binding was calculated using the following equation:

$$\text{Serum protein binding (\%)} = (Cs - Cb) / Cs \times 100$$

Cs and Cb represent the concentration of the test compound in the serum and buffer, respectively. The instrument used was API4000 mass spectrometer (AB SCIEX) equipped with a high-performance liquid chromatography system (Shimadzu Corp.).

4.6. In silico study

4.6.1. Fragment molecular orbital (FMO) calculation

The co-crystal structures of **2** and **4e** with HPSE1 (PDB code: 7YI7[21] and 8JYG) were analyzed using the FMO method implemented in GAMESS,[41,42] as follows. Making input and analysis of FMO calculations were performed using Scientific Vector Language source code "MOE_FMOutil.svl".[43] Molecular modeling was performed using MOE software.[44] Crystal water molecules within 4.5 Å from ligand, solvent molecules (ethylene glycol), and receptor atoms retained. ACE and NME capping groups were added to the N- and C-terminus, respectively. Hydrogen atoms were added to the complex structure using the MOE/QuickPrep tool, and their positions were optimized using Amber10: EHT force field parameters with the generalized Born implicit solvent model. Then, we performed all FMO calculations at the second order Møller–Plesset perturbation method based on the Resolution-of-Identity Approximation (RI-MP2) with the 6-31G* basis set.[45] The cc-pVDZ auxiliary basis set was applied for the procedure of RI approximation. In our FMO calculations, each amino acid unit containing side chain, Cα, and *N*-terminal side amino group, was assigned to one fragment. Each ligand was treated as single fragment. PIE values between ligands (**2** and **4e**) and each amino acid residue located within 4.5 Å from these ligands were calculated to reveal their energy contribution shown in Table S1–3.

4.6.2. Molecular superposition

The superpositions of **4e** and the X-ray co-crystal structure of a sugar analogue 2-propyl uronic isofagomine and GUSβ (PDB code: 6LEJ)[37] or that of a sugar analogue *N*-butyl-deoxynojirimycin and GBA (PDB code: 2V3D)[38], shown in Fig. 4**A and 4B**, were performed using MOE software.[44].

As illustrated in the superposition in Fig. 4**A**, the 5-carboxy group, 6-, 7-hydroxy groups, 8-hydroxy group, and 8a-carbon atom of **4e** correspond to the 3-carboxy group, 4-, 5-hydroxy groups, 6-methylene group, and 1-amino group of 2-propyl uronic isofagomine, respectively.

As illustrated in the superposition in Fig. 4**B**, the 5-carboxy group, 6-, 7-, 8-hydroxy groups, and 8a-carbon atom of **4e** correspond to the 2-hydroxymethyl group, 3-, 4-, 5-hydroxy groups, and 6-methylene group of *N*-butyl-deoxynojirimycin, respectively.

The visualization of the superpositions was performed using PyMOL software.[46].

4.7. X-ray co-crystal structures of **4e** and HPSE1

4.7.1. Protein production

cDNA fragments encoding residues 36–109 and 158–543 of human HPSE1 (AF155510) plus the honeybee melittin signal sequence were obtained from GeneArt™ (Thermo Fisher Scientific Inc., Waltham, MA, USA) and subcloned into pIEx4 (Novagen, Merck KGaA Ltd., Darmstadt, Germany) vectors respectively. High Five™ insect cells (Thermo Fisher Scientific) were grown in Express Five™ SFM medium (Thermo Fisher Scientific) and then, transfected with recombinant plasmids (36–109/pIEx4 and 158–543/pIEx4) using Escort™ IV transfection reagent (Sigma-Aldrich, Merck KGaA Ltd., Darmstadt, Germany), according to the manufacturer's instructions. The cells were incubated at 28 °C with shaking at 120 rpm for 72 h. The conditioned medium was clarified by centrifugation and purified using a HiTrap™ Heparin HP column (GE HealthCare, Chicago, IL, USA) and a HiTrap™ SP HP column (GE HealthCare). HPSE-containing fractions were concentrated and treated with endoglycosidase H (New England Biolabs Inc., Ipswich, MA, USA)

Table 4
X-ray Data Collection and Refinement Statistics.

Compound	4e
Data Collection	
space group	*P*21
Cell Dimensions	
a (Å)	44.38
b (Å)	71.05
c (Å)	78.28
α (deg)	90.00
β (deg)	98.09
γ (deg)	90.00
resolution (Å)	2.00 (2.05–2.00)
Rmerge (%)	13.5 (89.4)
I/σI	7.5 (2.0)
completeness (%)	99.8 (99.8)
redundancy	5.3 (5.4)
Refinement	
resolution (Å)	43.98–2.00
no. of reflections (work/free)	31005/1600
Rwork/Rfree (%)	16.6/20.9
rms deviation	
bond lengths (Å)	0.0089
bond angles (deg)	1.58

for 4 h at 25 °C. The digested protein was purified by size-exclusion chromatography (*SEC*) with a Superdex 75 Increase 10/300 GL column (GE HealthCare), with the buffer exchanged to 20 mM HEPES pH7.5, 100 mM NaCl, 1 mM DTT and concentrated to 10 mg/mL.

4.7.2. Crystallization, data collection, and processing

Complexes with **4e** were crystallized by vapor diffusion at 20 °C by mixing equal volumes of complexes with precipitant [0.1 M BIS-TRIS (pH5.5), 25 % PEG3350]. The crystals were flash-frozen at a temperature of 100 K using a cryoprotection buffer. Diffraction data were collected at the Swiss Light Source X10SA. Data processing was performed using iMosflm[47]. The structure was determined by molecular replacement with MolRep[48] using the structure with the PDB code 5E8M as the search model. The structure was refined with Refmac5.[49] The data collection and refinement statistics are summarized in Table 4.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bmc.2023.117460.

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